PAPER_886: Erosion Lagrangian Euler-Lagrange

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** ErosionLagrangianEulerLagrangeCalc (CP4 #470) **CVW:** v2.0.0 compliant

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Abstract

Symmetric counterpart to the expansion Lagrangian. L_erosion = $E^{-}(t) \cdot V \cdot S_{26}$ with Euler-Lagrange variation $\delta S/\delta\varphi$_erosion = 0 providing closure for the erosion potential in filaments and cavities where gravity dominates buoyancy.

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1. Core Equations

$$L\_erosion = E^{-}(t) \cdot V \cdot S_{26}$$
$$\delta S/\delta\varphi\_erosion = 0$$

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2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter
V_filament	1e48 m^3	Filament volume
F_UBi_over_FU	0.3	Buoyancy-to-field ratio

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3. UQFF Integration

This calculator integrates negative_et_erosion.py from Session 205 into the CP4 pipeline as class #470.

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